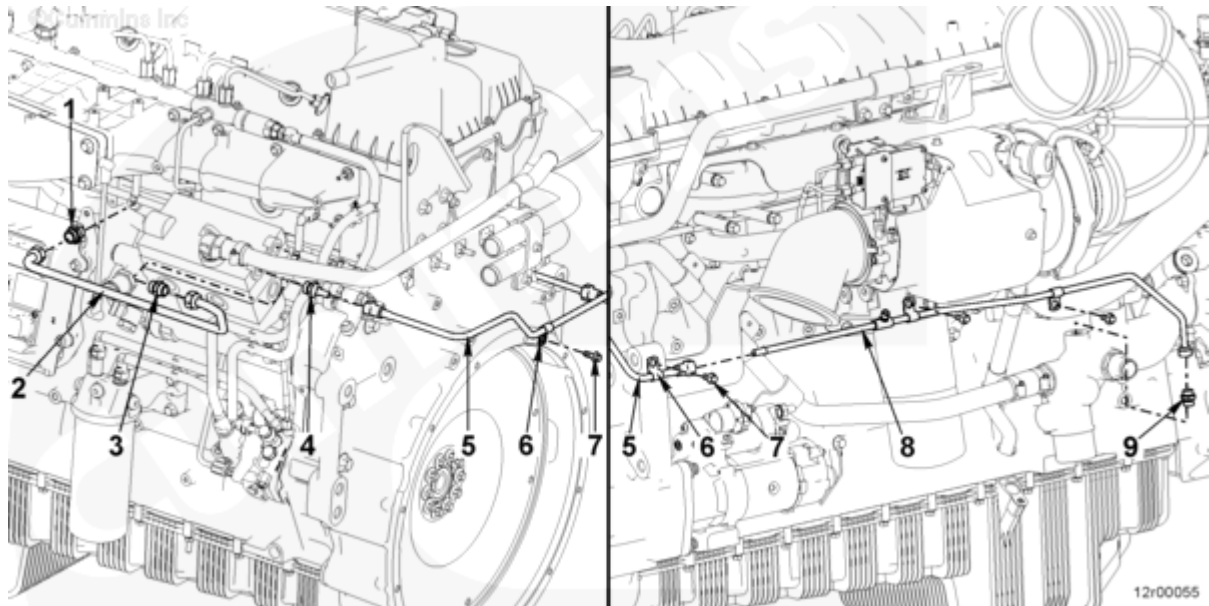


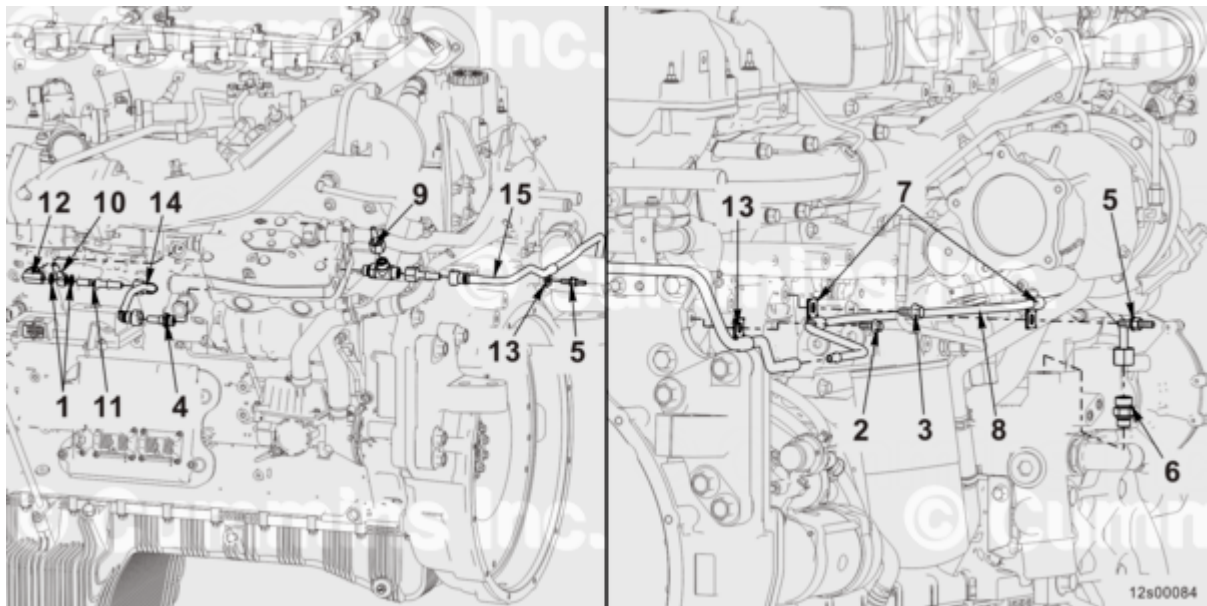
Trainings ▾

Exploded View



Air Compressor Coolant Lines (Without Water Jacket) Exploded View

1. Male union - quantity 1
2. Inlet tube - quantity 1
3. Male union - quantity 1
4. Male connector - quantity 1
5. Outlet tube - quantity 1
6. Clip - quantity 4
7. Capscrew - quantity 4
8. Outlet tube - quantity 1
9. Quick-disconnect connector - quantity 1



Air Compressor Coolant Lines (With Water Jacket) Exploded View

1. Sealing washer - quantity 2
2. Capscrew - quantity 1
3. Capscrew - quantity 1
4. Quick-disconnect connector - quantity 1
5. Stud - quantity 2
6. Male union - quantity 1
7. Clip - quantity 2
8. Outlet tube - quantity 1
9. Male union tee - quantity 1
10. Banjo connector - quantity 1
11. Banjo connector screw - quantity 1
12. Elbow - quantity 1
13. Clip - quantity 2
14. Inlet tube - quantity 1
15. Outlet tube - quantity 1

Select Service Tools

Recommended Cummins® Service Tools

- No recommended Cummins® service tools required

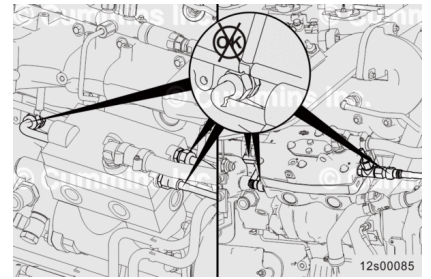
Additional Service Items

- No additional service items required.

Initial Check

Inspect metal coolant lines for kinks, corrosion, or cracks.

Replace any damaged components.



Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [122°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

Note : Draining the cooling system is **not** required on air-cooled air compressors.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html>)
- Disconnect the water transfer tube from air intake connection to the male union tee. Refer to Procedure 008-152 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-152.html>)

Remove

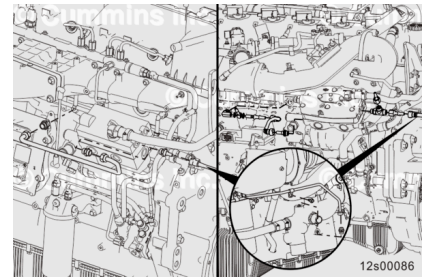
Remove four clip screws of coolant outlet tube from cylinder block.

Remove coolant outlet tube.

Remove coolant water inlet tube.

For the engines without Exhaust Gas Recirculation (EGR) water jacket, remove the male union, male connector, and quick-disconnect connectors, if required.

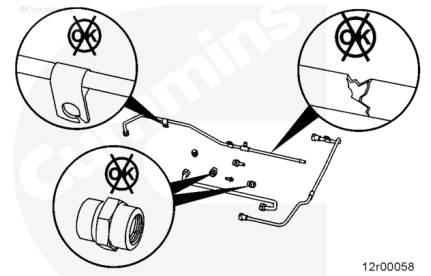
For the engines with EGR water jacket, remove the male union, male connector, banjo connector, male union tee, and quick-disconnect connectors, if required.



Inspect for Reuse

Inspect the drive clips or connectors for damaged threads.

Inspect coolant lines for restriction of coolant flow.



Install

⚠ CAUTION ⚠

If rubber grommets are used on the coolant lines, be sure they are installed carefully to prevent cuts or tears to the grommets, which can cause leaks.

For the engines without EGR water jacket:

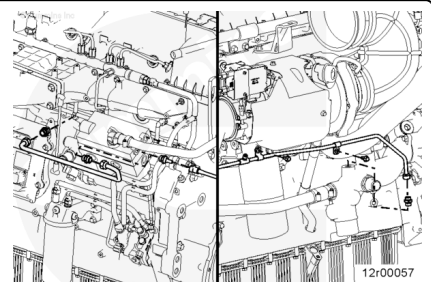
Install one male union to intake side of cylinder block, and install one male union to the front side of air compressor, if removed.

Torque Value: 61 N·m [45 ft-lb]

Install quick-disconnect connectors onto the front side of air compressor, and install male connector to the top of the coolant inlet pipe, if removed.

Torque Value: 61 N·m [45 ft-lb]

Install coolant water inlet tube. Tighten two fittings.



Torque Value: 27 n•m [239 in-lb]

Install coolant outlet tube. Tighten fittings.

Torque Value: 27 n•m [239 in-lb]

Install four clip screws of coolant outlet tube into cylinder block. Tighten screws.

Torque Value: 20 n•m [177 in-lb]

For the engines with EGR water jacket:

Install the elbow to intake side of cylinder block.

Torque Value: 61 n•m [45 ft-lb]

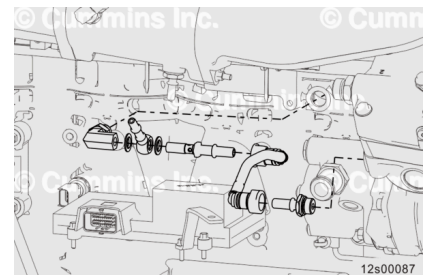
Install the quick-disconnect connector to air compressor.

Torque Value: 45 n•m [33 ft-lb]

Install two sealing washers, banjo connector, and banjo screw into the elbow. Verify the banjo connector is angled 45 degrees upward from horizontal.

Torque Value: 24 n•m [212 in-lb]

Install the inlet tube.



Install the male union tee to air compressor. Verify the quick-disconnect connector is perpendicular to the top surface of air compressor.

Torque Value: 45 n•m [33 ft-lb]

Connect the outlet tube to the male union tee. Install two clip screws of outlet tube into flywheel housing. Tighten screws.

Torque Value: 24 n•m [212 in-lb]

Install the outlet tube to water inlet connection. Connect the outlet tubes.

Torque Value: 27 n•m [239 in-lb]

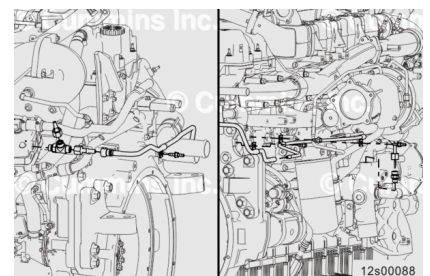
Install two clip screws of outlet tube into cylinder block. Tighten screws.

Torque Value:

M8 18 n•m [159 in-lb]

Torque Value:

M10 47 n•m [35 ft-lb]



Finishing Steps

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Connect the water transfer tube from air intake connection to the male union tee. Refer to Procedure 008-152 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-152.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-018.html>)
- Operate the engine to normal operating temperature and check for leaks.